CARE DASHBOARD SCENARIO ANALYSIS PAGE: MODEL ESTIMATION RESULTS AND ATE COMPUTATION METHODS

This document presents the model estimation results and average treatment effect (ATE) computation methods used on the CARE Dashboard's Scenario Analysis page. The model estimation team includes [Irfan Batur, Jinghai Huo, Miguel Rodriguez Ocana, Fan Yu, Victor Ojimaojo Alhassan, Roberto Dimas Valle, and Fawzan Saleh M Alfawzan]. For questions, please contact Dr. Irfan Batur at ibatur@asu.edu.

1. CONFIRMATORY FACTOR ANALYSIS

The CARE survey includes a range of attitudinal statements that were used to capture respondents' latent attitudes, personality traits, and lifestyle preferences. A confirmatory factor analysis groups nine of these statements into three latent constructs: social interaction propensity (SE), community support capacity (CR), and personal resilience (PR). Table 1 reports the factor loadings. The resulting factor scores, standardized to mean ≈ 0 and SD ≈ 1, enter the models in Section 2 as continuous explanatory variables.

TABLE 1 Factor Loadings of Latent Constructs on Attitudinal Indicators

Attitudinal Indicators	Loadings of Latent Constructs on Indicators						
	Social Interaction Propensity (SE)		Community Support Capacity (CR)		Personal Resilience (PR)		Std. loading
	Coef.	*z-stat*	*Coef.*	*z-stat*	*Coef.*	*z-stat*	
I enjoy the chance to meet people unexpectedly, interact with strangers, or make new acquaintances when I visit public places.	0.753	45.2	—	—	—	—	0.664
I feel energized when I am part of a large group.	0.831	47.8	—	—	—	—	0.700
I regularly participate in community activities, which strengthen my sense of belonging and support.	0.842	45.1	—	—	—	—	0.663
I believe my community would come together to support each other during an extreme event.	—	—	0.746	46.8	—	—	0.698
I have family or friends nearby that I can turn to when I need immediate help.	—	—	0.633	33.4	—	—	0.512
My community has effective communication channels to distribute information during an extreme event.	—	—	0.779	46.8	—	—	0.697
I feel that I can personally bounce back quickly from challenging situations.	—	—	—	—	0.643	45.7	0.668
I am knowledgeable about the steps I need to take for my safety during extreme events.	—	—	—	—	0.592	42.0	0.618
[Item wording for att_adaptability to be inserted]	—	—	—	—	0.722	52.2	0.759

Note: Coef. = unstandardized loading; Std. loading = standardized loading; "—" = not applicable (indicator not assigned to that construct). All loadings are significant at $p < 0.001$.

2. MODEL ESTIMATION RESULTS

This section reports, for each of nine activities, how respondents' reported behavior during five extreme events (extreme heat, extreme cold, power outage, earthquake, and flooding) is distributed, followed by the ordered probit models that relate these responses to the self-reported severity of the event's impact, the latent constructs from Section 1, and individual, household, and built-environment characteristics. A separate model is estimated for each event × activity combination (35 models). Working from home and working from the office are modeled for workers only, and an event is omitted from an activity's tables when that activity was not asked for the event.

2.1. GOING ABOUT BUSINESS AS USUAL

TABLE 2 Distribution of Responses across Events — Going about business as usual

		Likelihood of going about business as usual — % of respondents (n)				
Event	**N**	**Very unlikely**	**Somewhat unlikely**	**Neutral**	**Somewhat likely**	**Very likely**
Extreme Heat	2,653	7.8 (208)	14.4 (382)	19.5 (517)	34.8 (924)	23.4 (622)
Extreme Cold	2,434	5.8 (142)	15.1 (367)	20.7 (505)	34.8 (846)	23.6 (574)
Power Outage	2,381	10.5 (251)	19.3 (459)	22.8 (544)	31.1 (741)	16.2 (386)
Earthquake	611	11.0 (67)	17.7 (108)	24.5 (150)	30.3 (185)	16.5 (101)
Flooding	1,077	11.5 (124)	18.4 (198)	24.2 (261)	30.5 (328)	15.4 (166)

Note: Cells show the share of respondents in each response category, % (n). All respondents exposed to each event; N = number of respondents.

TABLE 3 Model Estimation Results across Events — Going about business as usual

Variable (base)	Attribute	Ordered probit — outcome: likelihood of going about business as usual (1 = Very unlikely, 2 = Somewhat unlikely, 3 = Neutral, 4 = Somewhat likely, 5 = Very likely)									
		Extreme Heat		Extreme Cold		Power Outage		Earthquake		Flooding	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact											
Impact severity (Not severe at all)	Slightly severe	-0.659	-9.624	-0.584	-8.885	-0.419	-6.975	-0.268	-2.281	-0.476	-5.477
	Moderately severe	-0.908	-13.252	-0.800	-11.870	-0.538	-8.701	-0.353	-3.042	-0.537	-5.869
	Very severe	-1.089	-13.289	-0.909	-11.318	-0.768	-9.703	-0.187	-1.323	-0.664	-5.735
	Extremely severe	-1.032	-10.840	-0.970	-9.520	-0.917	-9.135	-0.055	-0.275	-0.617	-4.328
Latent Constructs											
Factor score (continuous)	Social interaction propensity	--	--	--	--	0.099	3.535	--	--	--	--
	Community support capacity	0.082	3.114	0.089	3.402	0.084	3.119	--	--	0.085	2.167
	Personal resilience	0.182	6.837	0.194	7.063	0.087	3.141	0.176	3.194	0.109	2.702
Individual Characteristics											
Gender (Male)	Female	-0.111	-2.525	-0.139	-3.098	-0.119	-2.578	-0.274	-3.073	-0.175	-2.638
Age (*)	31 to 50 years	0.188	3.617	0.101	2.088	--	--	--	--	0.204	2.942
	51 to 65 years	0.109	2.027	--	--	-0.121	-2.179	-0.438	-3.778	--	--
	65 or older	--	--	--	--	-0.307	-5.693	-0.483	-4.478	--	--
Race (*)	White	--	--	0.206	3.721	--	--	0.186	1.784	0.180	2.225
Employment (Worker)	Non-worker	-0.206	-4.386	-0.224	-4.727	--	--	--	--	-0.166	-2.081
Disability (No)	Travel-related disability	-0.128	-2.384	--	--	--	--	--	--	--	--
Work setting (Other)	Works mainly outdoors	0.278	2.619	--	--	0.315	2.921	--	--	--	--
Telecommuting (Yes)	Does not telecommute	--	--	--	--	--	--	-0.202	-1.778	-0.338	-3.873
Household Characteristics											
Income (*)	Under $50,000	--	--	--	--	-0.133	-2.296	--	--	--	--
	$50,000 to $100,000	--	--	--	--	-0.160	-2.647	--	--	--	--
Vehicles (1 or more)	Zero vehicles	-0.141	-2.152	--	--	--	--	--	--	--	--
Dwelling (Other)	Stand-alone house	--	--	-0.080	-1.812	--	--	--	--	--	--
Air conditioning (No)	Has A/C	0.233	3.825	0.126	2.814	--	--	--	--	--	--
Built Environment											
Area type (Non-rural)	Rural	--	--	--	--	0.162	1.998	0.451	2.058	--	--
Population density, people/acre (*)	Medium (>4.00–8.86)	0.082	1.851	--	--	--	--	--	--	--	--
	High (>8.86)	--	--	--	--	0.077	1.606	--	--	--	--
Thresholds											
Cut-points	τ1 (1 | 2)	-2.153	-21.906	-2.283	-24.273	-2.013	-24.882	-1.727	-12.469	-1.679	-13.913
	τ2 (2 | 3)	-1.452	-15.328	-1.474	-16.752	-1.247	-16.370	-1.031	-7.921	-0.986	-8.486
	τ3 (3 | 4)	-0.848	-9.099	-0.831	-9.654	-0.618	-8.303	-0.337	-2.680	-0.322	-2.827
	τ4 (4 | 5)	0.181	1.966	0.178	2.093	0.351	4.732	0.618	4.858	0.647	5.644
Model Fit											
Number of observations		2,653		2,434		2,381		611		1,077	
Log-likelihood at convergence		-3775.614		-3451.472		-3570.943		-914.808		-1627.973	
Log-likelihood (thresholds only)		-3992.168		-3615.321		-3690.699		-948.751		-1673.714	
McFadden pseudo-R^2		0.054		0.045		0.032		0.036		0.027	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Built-environment categories: population density (people/acre): low ≤4.00, medium >4.00–8.86, high >8.86. Sample: all respondents exposed to each event.

2.2. STAYING AT HOME

TABLE 4 Distribution of Responses across Events — Staying at home

Event	N	Change in staying at home — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	2,653	6.2 (165)	49.7 (1,318)	44.1 (1,170)	+37.9
Extreme Cold	2,434	4.1 (101)	42.4 (1,032)	53.5 (1,301)	+49.3
Power Outage	2,381	12.1 (288)	51.5 (1,227)	36.4 (866)	+24.3
Earthquake	611	7.5 (46)	48.8 (298)	43.7 (267)	+36.2
Flooding	1,077	5.9 (64)	46.1 (496)	48.0 (517)	+42.1

Note: Cells show the share of respondents in each response category, % (n). All respondents exposed to each event; N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 5 Model Estimation Results across Events — Staying at home

Variable (base)	Attribute	Ordered probit — outcome: change in staying at home (1 = Less, 2 = About the same, 3 = More)									
		Extreme Heat		Extreme Cold		Power Outage		Earthquake		Flooding	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact											
Impact severity (Not severe at all)	Slightly severe	0.272	3.849	0.298	4.257	0.149	2.316	0.286	2.225	0.218	2.288
	Moderately severe	0.291	4.134	0.457	6.326	0.169	2.570	0.276	2.167	0.201	2.030
	Very severe	0.479	5.515	0.334	3.887	0.098	1.171	0.267	1.752	0.088	0.707
	Extremely severe	0.387	3.808	0.430	3.841	0.268	2.531	0.358	1.649	0.253	1.622
Latent Constructs											
Factor score (continuous)	Social interaction propensity	--	--	-0.079	-2.643	--	--	--	--	--	--
	Community support capacity	0.086	3.093	0.076	2.647	--	--	--	--	0.118	2.767
	Personal resilience	--	--	--	--	0.076	2.570	--	--	--	--
Individual Characteristics											
Gender (Male)	Female	--	--	--	--	--	--	--	--	0.169	2.314
Age (*)	51 to 65 years	--	--	--	--	0.184	3.085	--	--	--	--
	65 or older	--	--	--	--	0.243	4.233	--	--	--	--
Education (*)	Bachelor's degree	0.104	1.890	--	--	--	--	--	--	--	--
	Bachelor's or higher	--	--	--	--	--	--	--	--	0.195	2.486
Race (*)	White	--	--	--	--	--	--	0.189	1.587	--	--
	Black	--	--	--	--	0.160	2.266	--	--	--	--
	Asian	--	--	--	--	--	--	0.565	2.480	--	--
Ethnicity (Non-Hispanic)	Hispanic	0.159	1.821	--	--	--	--	--	--	--	--
Employment (Worker)	Non-worker	0.159	2.781	--	--	--	--	0.222	2.222	--	--
Disability (No)	Travel-related disability	0.147	2.485	0.113	1.762	0.160	2.600	--	--	--	--
Work setting (Other)	Works mainly outdoors	-0.237	-2.105	--	--	--	--	--	--	--	--
Telecommuting (Yes)	Does not telecommute	0.124	1.920	-0.099	-1.740	--	--	--	--	--	--
Household Characteristics											
Income (*)	$50,000 to $100,000	--	--	--	--	-0.097	-1.916	--	--	--	--
Household size (3+)	2 persons	-0.084	-1.742	--	--	--	--	--	--	--	--
Air conditioning (No)	Has A/C	0.176	2.649	--	--	--	--	--	--	--	--
Built Environment											
Population density, people/acre (*)	High (>8.86)	0.146	2.960	--	--	--	--	--	--	--	--
Employment density, jobs/acre (*)	High (>2.37)	--	--	--	--	0.092	1.814	--	--	--	--
Transit access (*)	High (>0.13)	--	--	--	--	--	--	--	--	0.209	1.953
Thresholds											
Cut-points	τ1 (Less | Same)	-0.955	-9.603	-1.457	-21.496	-0.894	-13.188	-1.001	-7.163	-1.260	-13.076
	τ2 (Same | More)	0.764	7.784	0.214	3.662	0.646	9.696	0.629	4.653	0.378	4.315
Model Fit											
Number of observations		2,653		2,434		2,381		611		1,077	
Log-likelihood at convergence		-2293.721		-1991.616		-2271.689		-544.258		-928.575	
Log-likelihood (thresholds only)		-2338.193		-2021.844		-2297.649		-553.977		-944.675	
McFadden pseudo-R^2		0.019		0.015		0.011		0.018		0.017	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Built-environment categories: population density (people/acre): low ≤4.00, medium >4.00–8.86, high >8.86; employment density (jobs/acre): low ≤0.91, medium >0.91–2.37, high >2.37; transit access: low = 0, medium >0–0.13, high >0.13. Sample: all respondents exposed to each event.

2.3. USING A CAR FOR TRAVELING

TABLE 6 Distribution of Responses across Events — Using a car for traveling

Event	N	Change in using a car for traveling — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	2,653	18.5 (492)	64.2 (1,703)	17.3 (458)	-1.3
Extreme Cold	2,434	30.4 (741)	55.2 (1,343)	14.4 (350)	-16.1
Power Outage	2,381	17.9 (426)	61.4 (1,462)	20.7 (493)	+2.8
Earthquake	611	30.9 (189)	52.5 (321)	16.5 (101)	-14.4
Flooding	1,077	39.9 (430)	48.1 (518)	12.0 (129)	-27.9

Note: Cells show the share of respondents in each response category, % (n). All respondents exposed to each event; N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 7 Model Estimation Results across Events — Using a car for traveling

Variable (base)	Attribute	Ordered probit — outcome: change in using a car for traveling (1 = Less, 2 = About the same, 3 = More)									
		Extreme Heat		Extreme Cold		Power Outage		Earthquake		Flooding	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact											
Impact severity (Not severe at all)	Slightly severe	0.053	0.749	-0.060	-0.878	-0.011	-0.169	-0.213	-1.677	-0.065	-0.690
	Moderately severe	0.128	1.806	-0.056	-0.804	0.066	0.998	0.078	0.629	-0.034	-0.338
	Very severe	0.157	1.854	0.021	0.253	0.089	1.055	0.331	2.206	0.076	0.608
	Extremely severe	0.167	1.696	0.188	1.754	0.196	1.856	0.204	0.971	0.265	1.760
Latent Constructs											
Factor score (continuous)	Social interaction propensity	0.076	2.724	0.082	2.848	--	--	--	--	0.142	3.180
	Personal resilience	0.075	2.663	--	--	--	--	0.117	1.993	-0.117	-2.723
Individual Characteristics											
Gender (Male)	Female	--	--	-0.082	-1.683	-0.170	-3.520	-0.222	-2.320	--	--
Age (*)	51 to 65 years	--	--	-0.131	-2.129	-0.229	-3.603	-0.374	-3.006	--	--
	65 or older	--	--	-0.149	-2.235	-0.266	-4.251	-0.497	-4.370	-0.266	-2.964
Race (*)	White	0.174	1.941	--	--	--	--	--	--	--	--
	Black	0.246	2.399	0.208	3.005	--	--	--	--	--	--
Ethnicity (Non-Hispanic)	Hispanic	0.156	1.817	--	--	--	--	--	--	--	--
Employment (Worker)	Non-worker	-0.160	-3.377	-0.125	-2.244	--	--	--	--	--	--
Work setting (Other)	Works mainly outdoors	--	--	--	--	--	--	--	--	0.608	4.035
Household Characteristics											
Income (*)	$50,000 to $100,000	--	--	-0.097	-1.877	--	--	--	--	--	--
Household size (3+)	1 person	--	--	--	--	--	--	0.222	2.043	--	--
	2 persons	-0.115	-2.399	--	--	--	--	--	--	--	--
Children (None)	Has child(ren)	--	--	--	--	0.128	2.082	--	--	0.128	1.551
Vehicles (1 or more)	Zero vehicles	--	--	--	--	-0.321	-4.390	--	--	0.232	2.195
Dwelling (Other)	Stand-alone house	-0.083	-1.777	-0.109	-2.246	--	--	--	--	--	--
Built Environment											
Population density, people/acre (*)	High (>8.86)	0.088	1.811	0.153	2.978	--	--	--	--	--	--
Transit access (*)	High (>0.13)	--	--	--	--	--	--	--	--	-0.183	-1.764
Thresholds											
Cut-points	τ1 (Less | Same)	-0.770	-6.830	-0.745	-9.053	-1.150	-15.802	-0.776	-6.802	-0.237	-2.820
	τ2 (Same | More)	1.105	9.725	0.882	10.611	0.631	8.931	0.794	6.911	1.264	13.806
Model Fit											
Number of observations		2,653		2,434		2,381		611		1,077	
Log-likelihood at convergence		-2348.145		-2301.784		-2174.309		-580.951		-1011.672	
Log-likelihood (thresholds only)		-2388.456		-2358.628		-2222.476		-610.174		-1047.713	
McFadden pseudo-R^2		0.017		0.024		0.022		0.048		0.034	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Built-environment categories: population density (people/acre): low ≤4.00, medium >4.00–8.86, high >8.86; transit access: low = 0, medium >0–0.13, high >0.13. Sample: all respondents exposed to each event.

2.4. TAKING PUBLIC TRANSIT

TABLE 8 Distribution of Responses across Events — Taking public transit

Event	N	Change in taking public transit — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	2,653	23.1 (614)	68.0 (1,803)	8.9 (236)	-14.2
Extreme Cold	2,434	26.5 (645)	64.5 (1,570)	9.0 (219)	-17.5
Power Outage	2,381	22.0 (524)	68.1 (1,622)	9.9 (235)	-12.1
Earthquake	611	23.6 (144)	60.2 (368)	16.2 (99)	-7.4
Flooding	1,077	23.3 (251)	63.6 (685)	13.1 (141)	-10.2

Note: Cells show the share of respondents in each response category, % (n). All respondents exposed to each event; N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 9 Model Estimation Results across Events — Taking public transit

Variable (base)	Attribute	Ordered probit — outcome: change in taking public transit (1 = Less, 2 = About the same, 3 = More)									
		Extreme Heat		Extreme Cold		Power Outage		Earthquake		Flooding	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact											
Impact severity (Not severe at all)	Slightly severe	-0.089	-1.214	-0.081	-1.144	-0.088	-1.305	0.073	0.565	-0.058	-0.617
	Moderately severe	-0.141	-1.932	-0.170	-2.365	-0.054	-0.782	0.220	1.724	0.001	0.006
	Very severe	-0.106	-1.213	-0.187	-2.172	-0.015	-0.167	0.352	2.232	0.087	0.697
	Extremely severe	-0.146	-1.436	0.129	1.184	0.156	1.432	0.217	0.992	-0.023	-0.151
Latent Constructs											
Factor score (continuous)	Social interaction propensity	--	--	--	--	0.057	1.791	-0.126	-1.946	--	--
	Community support capacity	--	--	0.089	3.128	0.069	2.272	--	--	0.079	1.879
	Personal resilience	-0.060	-2.071	--	--	-0.053	-1.692	--	--	--	--
Individual Characteristics											
Gender (Male)	Female	--	--	--	--	-0.098	-1.895	-0.198	-1.998	--	--
Age (*)	31 to 50 years	-0.156	-2.076	--	--	-0.219	-2.658	--	--	-0.142	-1.824
	51 to 65 years	-0.153	-1.921	--	--	-0.265	-2.965	--	--	--	--
	65 or older	-0.157	-1.961	--	--	-0.272	-3.039	--	--	--	--
Education (*)	Bachelor's degree	--	--	--	--	--	--	--	--	0.254	2.895
	Bachelor's or higher	--	--	--	--	--	--	-0.207	-2.099	--	--
Race (*)	Asian	--	--	-0.244	-1.731	--	--	--	--	--	--
Ethnicity (Non-Hispanic)	Hispanic	--	--	--	--	-0.209	-2.207	0.371	2.483	--	--
Employment (Worker)	Non-worker	--	--	--	--	--	--	-0.271	-2.308	--	--
Work setting (Other)	Works mainly outdoors	0.465	4.102	0.281	2.370	--	--	--	--	--	--
Telecommuting (Yes)	Does not telecommute	-0.160	-2.822	--	--	--	--	-0.322	-2.386	--	--
Household Characteristics											
Income (*)	Under $50,000	--	--	-0.188	-2.825	-0.167	-2.436	--	--	--	--
	$50,000 to $100,000	0.113	2.228	-0.189	-2.739	-0.156	-2.289	--	--	--	--
Household size (3+)	1 person	--	--	--	--	--	--	0.297	2.708	--	--
	2 persons	--	--	-0.107	-2.004	--	--	--	--	--	--
Children (None)	Has child(ren)	--	--	-0.174	-2.925	0.099	1.526	--	--	0.163	2.000
Vehicles (1 or more)	Zero vehicles	0.313	4.312	0.176	2.286	0.484	6.074	--	--	--	--
Dwelling (Other)	Stand-alone house	--	--	-0.152	-2.945	-0.090	-1.722	--	--	--	--
Air conditioning (No)	Has A/C	--	--	--	--	--	--	--	--	0.154	1.952
Built Environment											
Transit access (*)	High (>0.13)	-0.126	-1.728	-0.189	-2.500	--	--	--	--	--	--
Walkability index (*)	Very high (>13)	--	--	--	--	--	--	--	--	0.229	2.895
Thresholds											
Cut-points	τ1 (Less | Same)	-0.942	-10.059	-1.046	-11.341	-1.215	-10.926	-0.882	-6.289	-0.596	-7.107
	τ2 (Same | More)	1.170	12.337	0.959	10.375	0.910	8.257	0.898	6.383	1.297	14.342
Model Fit											
Number of observations		2,653		2,434		2,381		611		1,077	
Log-likelihood at convergence		-2138.549		-2038.032		-1913.536		-555.612		-944.796	
Log-likelihood (thresholds only)		-2165.977		-2072.369		-1960.032		-574.880		-962.228	
McFadden pseudo-R^2		0.013		0.017		0.024		0.034		0.018	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Built-environment categories: transit access: low = 0, medium >0–0.13, high >0.13; National Walkability Index: very low ≤8, low >8–10, high >10–13, very high >13. Sample: all respondents exposed to each event.

2.5. WORKING FROM HOME

TABLE 10 Distribution of Responses across Events — Working from home

Event	N	Change in working from home — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	1,271	7.4 (94)	66.5 (845)	26.1 (332)	+18.7
Extreme Cold	1,179	7.1 (84)	59.5 (701)	33.4 (394)	+26.3
Earthquake	343	10.5 (36)	60.1 (206)	29.4 (101)	+19.0
Flooding	585	9.4 (55)	61.4 (359)	29.2 (171)	+19.8

Note: Cells show the share of respondents in each response category, % (n). Workers only (currently employed respondents exposed to each event); N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 11 Model Estimation Results across Events — Working from home

Variable (base)	Attribute	Ordered probit — outcome: change in working from home (1 = Less, 2 = About the same, 3 = More)							
		Extreme Heat		Extreme Cold		Earthquake		Flooding	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact									
Impact severity (Not severe at all)	Slightly severe	0.001	0.006	0.053	0.477	0.021	0.112	0.148	1.064
	Moderately severe	0.026	0.237	0.241	2.160	0.365	2.102	0.185	1.320
	Very severe	0.336	2.628	0.036	0.288	0.183	0.908	0.159	0.942
	Extremely severe	0.068	0.461	0.203	1.311	0.047	0.189	0.521	2.440
Latent Constructs									
Factor score (continuous)	Community support capacity	0.121	2.785	--	--	--	--	0.131	2.208
Individual Characteristics									
Gender (Male)	Female	--	--	-0.150	-2.147	--	--	--	--
Education (*)	Bachelor's degree	0.155	1.998	--	--	--	--	--	--
	Bachelor's or higher	--	--	0.203	2.802	--	--	0.205	2.014
Race (*)	White	--	--	-0.414	-2.742	--	--	--	--
	Black	0.216	2.429	-0.321	-1.975	--	--	--	--
	Asian	--	--	-0.498	-2.337	--	--	--	--
Ethnicity (Non-Hispanic)	Hispanic	0.288	2.600	--	--	--	--	--	--
Telecommuting (Yes)	Does not telecommute	-0.433	-6.226	-0.452	-6.296	-0.482	-3.512	-0.150	-1.492
Household Characteristics									
Income (*)	Under $50,000	-0.177	-1.995	--	--	--	--	--	--
	$50,000 to $100,000	-0.150	-1.704	0.116	1.590	--	--	--	--
Household size (3+)	2 persons	-0.172	-2.273	--	--	--	--	--	--
Children (None)	Has child(ren)	--	--	0.205	2.828	--	--	--	--
Air conditioning (No)	Has A/C	0.206	2.004	--	--	--	--	--	--
Thresholds									
Cut-points	τ1 (Less | Same)	-1.542	-9.770	-1.914	-9.730	-1.311	-8.301	-1.157	-8.358
	τ2 (Same | More)	0.662	4.311	0.079	0.415	0.545	3.733	0.750	5.616
Model Fit									
Number of observations		1,271		1,179		343		585	
Log-likelihood at convergence		-980.113		-968.038		-300.414		-503.432	
Log-likelihood (thresholds only)		-1035.434		-1018.207		-309.665		-515.652	
McFadden pseudo-R^2		0.053		0.049		0.030		0.024	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Sample: workers only (currently employed respondents exposed to each event).

2.6. WORKING FROM THE OFFICE

TABLE 12 Distribution of Responses across Events — Working from the office

Event	N	Change in working from the office — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	1,271	19.7 (251)	66.0 (839)	14.2 (181)	-5.5
Extreme Cold	1,179	27.8 (328)	58.7 (692)	13.5 (159)	-14.3

Note: Cells show the share of respondents in each response category, % (n). Workers only (currently employed respondents exposed to each event); N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 13 Model Estimation Results across Events — Working from the office

Variable (base)	Attribute	Ordered probit — outcome: change in working from the office (1 = Less, 2 = About the same, 3 = More)			
		Extreme Heat		Extreme Cold	
		Coef	*t-stat*	*Coef*	*t-stat*
Event Impact					
Impact severity (Not severe at all)	Slightly severe	-0.031	-0.276	-0.045	-0.417
	Moderately severe	-0.034	-0.319	-0.176	-1.637
	Very severe	-0.083	-0.675	-0.085	-0.703
	Extremely severe	0.318	2.235	0.269	1.800
Latent Constructs					
Factor score (continuous)	Community support capacity	-0.083	-1.998	--	--
Individual Characteristics					
Age (*)	31 to 50 years	--	--	-0.136	-2.034
Education (*)	Bachelor's degree	-0.204	-2.733	--	--
	Bachelor's or higher	--	--	-0.200	-2.933
Race (*)	White	--	--	0.454	3.013
	Black	--	--	0.574	3.536
	Asian	--	--	0.378	1.799
Ethnicity (Non-Hispanic)	Hispanic	--	--	0.244	2.207
Work setting (Other)	Works mainly outdoors	0.217	1.908	0.276	2.328
Telecommuting (Yes)	Does not telecommute	0.180	2.723	--	--
Household Characteristics					
Income (*)	Under $50,000	-0.117	-1.656	--	--
Household size (3+)	1 person	0.179	2.265	--	--
Thresholds					
Cut-points	τ1 (Less | Same)	-0.825	-7.605	-0.309	-1.742
	τ2 (Same | More)	1.134	10.268	1.429	7.867
Model Fit					
Number of observations		1,271		1,179	
Log-likelihood at convergence		-1088.307		-1082.336	
Log-likelihood (thresholds only)		-1108.406		-1106.928	
McFadden pseudo-R^2		0.018		0.022	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Sample: workers only (currently employed respondents exposed to each event).

2.7. EATING INDOORS AT A RESTAURANT

TABLE 14 Distribution of Responses across Events — Eating indoors at a restaurant

Event	N	Change in eating indoors at a restaurant — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	2,653	22.3 (592)	61.0 (1,618)	16.7 (443)	-5.6
Extreme Cold	2,434	36.6 (892)	51.2 (1,246)	12.2 (296)	-24.5
Power Outage	2,381	30.2 (719)	52.5 (1,250)	17.3 (412)	-12.9

Note: Cells show the share of respondents in each response category, % (n). All respondents exposed to each event; N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 15 Model Estimation Results across Events — Eating indoors at a restaurant

Variable (base)	Attribute	Ordered probit — outcome: change in eating indoors at a restaurant (1 = Less, 2 = About the same, 3 = More)					
		Extreme Heat		Extreme Cold		Power Outage	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact							
Impact severity (Not severe at all)	Slightly severe	0.101	1.449	-0.137	-2.030	0.016	0.256
	Moderately severe	0.005	0.079	-0.225	-3.235	0.118	1.810
	Very severe	-0.092	-1.100	-0.151	-1.808	0.271	3.293
	Extremely severe	0.259	2.670	0.128	1.209	0.255	2.464
Latent Constructs							
Factor score (continuous)	Social interaction propensity	--	--	0.086	2.973	0.056	1.895
	Personal resilience	--	--	--	--	-0.058	-1.961
Individual Characteristics							
Gender (Male)	Female	--	--	-0.123	-2.557	-0.117	-2.403
Age (*)	31 to 50 years	--	--	0.103	2.016	--	--
	51 to 65 years	--	--	--	--	-0.096	-1.773
Education (*)	Bachelor's degree	-0.125	-2.349	--	--	--	--
Race (*)	White	--	--	--	--	0.194	3.200
Employment (Worker)	Non-worker	-0.232	-5.131	-0.167	-3.254	--	--
Household Characteristics							
Income (*)	Under $50,000	--	--	--	--	-0.182	-2.944
	$50,000 to $100,000	--	--	-0.227	-4.403	-0.141	-2.201
Household size (3+)	2 persons	-0.112	-2.373	--	--	-0.094	-1.928
Built Environment							
Population density, people/acre (*)	High (>8.86)	--	--	0.095	1.885	--	--
Street network density, miles/sq. mile (*)	Medium (>0.40–1.11)	--	--	--	--	-0.144	-2.588
	High (>1.11)	--	--	--	--	-0.097	-1.712
Land-use diversity (*)	High (>0.45)	-0.137	-2.861	--	--	--	--
Walkability index (*)	High (>10–13)	--	--	0.151	2.635	--	--
Thresholds							
Cut-points	τ1 (Less | Same)	-0.957	-13.568	-0.599	-7.835	-0.619	-6.643
	τ2 (Same | More)	0.798	11.387	0.951	12.199	0.871	9.273
Model Fit							
Number of observations		2,653		2,434		2,381	
Log-likelihood at convergence		-2449.609		-2305.035		-2353.543	
Log-likelihood (thresholds only)		-2480.981		-2353.382		-2389.164	
McFadden pseudo-R^2		0.013		0.021		0.015	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Built-environment categories: population density (people/acre): low ≤4.00, medium >4.00–8.86, high >8.86; street network density (miles/sq. mile): low ≤0.40, medium >0.40–1.11, high >1.11; land-use diversity: low ≤0.34, medium >0.34–0.45, high >0.45; National Walkability Index: very low ≤8, low >8–10, high >10–13, very high >13. Sample: all respondents exposed to each event.

2.8. PICKING UP TAKEOUT

TABLE 16 Distribution of Responses across Events — Picking up takeout

Event	N	Change in picking up takeout — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	2,653	22.7 (601)	60.9 (1,615)	16.5 (437)	-6.2
Extreme Cold	2,434	34.1 (830)	51.6 (1,256)	14.3 (348)	-19.8
Power Outage	2,381	24.7 (589)	53.9 (1,283)	21.4 (509)	-3.4

Note: Cells show the share of respondents in each response category, % (n). All respondents exposed to each event; N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 17 Model Estimation Results across Events — Picking up takeout

Variable (base)	Attribute	Ordered probit — outcome: change in picking up takeout (1 = Less, 2 = About the same, 3 = More)					
		Extreme Heat		Extreme Cold		Power Outage	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact							
Impact severity (Not severe at all)	Slightly severe	-0.049	-0.699	-0.015	-0.224	-0.044	-0.701
	Moderately severe	0.017	0.240	-0.099	-1.433	0.044	0.679
	Very severe	0.054	0.647	-0.045	-0.546	0.158	1.915
	Extremely severe	0.030	0.303	0.105	0.996	0.093	0.900
Latent Constructs							
Factor score (continuous)	Social interaction propensity	0.082	2.970	0.080	2.795	--	--
	Community support capacity	--	--	--	--	0.048	1.721
	Personal resilience	0.049	1.788	0.091	3.144	--	--
Individual Characteristics							
Gender (Male)	Female	--	--	-0.158	-3.295	-0.124	-2.595
Age (*)	31 to 50 years	--	--	--	--	0.088	1.650
	51 to 65 years	-0.150	-2.648	--	--	--	--
	65 or older	-0.179	-3.328	--	--	--	--
Race (*)	White	--	--	--	--	0.114	1.772
	Asian	--	--	--	--	0.322	2.252
Employment (Worker)	Non-worker	--	--	-0.140	-2.857	-0.124	-2.450
Household Characteristics							
Income (*)	Under $50,000	--	--	--	--	-0.206	-3.334
	$50,000 to $100,000	--	--	-0.166	-3.242	-0.186	-2.889
Vehicles (1 or more)	Zero vehicles	-0.138	-1.965	--	--	--	--
Air conditioning (No)	Has A/C	-0.123	-1.892	--	--	--	--
Built Environment							
Population density, people/acre (*)	High (>8.86)	--	--	0.093	1.845	--	--
Employment density, jobs/acre (*)	Medium (>0.91–2.37)	-0.099	-2.086	--	--	--	--
Street network density, miles/sq. mile (*)	Medium (>0.40–1.11)	--	--	--	--	-0.127	-2.571
Land-use diversity (*)	High (>0.45)	-0.131	-2.723	--	--	--	--
Transit access (*)	High (>0.13)	--	--	--	--	-0.155	-1.975
Thresholds							
Cut-points	τ1 (Less | Same)	-1.049	-11.414	-0.633	-8.722	-0.890	-9.263
	τ2 (Same | More)	0.702	7.707	0.875	11.876	0.615	6.443
Model Fit							
Number of observations		2,653		2,434		2,381	
Log-likelihood at convergence		-2452.784		-2363.612		-2364.475	
Log-likelihood (thresholds only)		-2482.146		-2400.834		-2401.348	
McFadden pseudo-R^2		0.012		0.016		0.015	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Built-environment categories: population density (people/acre): low ≤4.00, medium >4.00–8.86, high >8.86; employment density (jobs/acre): low ≤0.91, medium >0.91–2.37, high >2.37; street network density (miles/sq. mile): low ≤0.40, medium >0.40–1.11, high >1.11; land-use diversity: low ≤0.34, medium >0.34–0.45, high >0.45; transit access: low = 0, medium >0–0.13, high >0.13. Sample: all respondents exposed to each event.

2.9. HAVING FOOD DELIVERED

TABLE 18 Distribution of Responses across Events — Having food delivered

Event	N	Change in having food delivered — % of respondents (n)			Net change, pp (More − Less)
		Less	About the same	More	
Extreme Heat	2,653	13.8 (366)	63.4 (1,683)	22.8 (604)	+9.0
Extreme Cold	2,434	20.4 (497)	56.0 (1,363)	23.6 (574)	+3.2
Power Outage	2,381	23.5 (559)	59.0 (1,404)	17.6 (418)	-5.9

Note: Cells show the share of respondents in each response category, % (n). All respondents exposed to each event; N = number of respondents. Net change = % More − % Less, in percentage points.

TABLE 19 Model Estimation Results across Events — Having food delivered

Variable (base)	Attribute	Ordered probit — outcome: change in having food delivered (1 = Less, 2 = About the same, 3 = More)					
		Extreme Heat		Extreme Cold		Power Outage	
		Coef	*t-stat*	*Coef*	*t-stat*	*Coef*	*t-stat*
Event Impact							
Impact severity (Not severe at all)	Slightly severe	0.156	2.184	0.009	0.137	-0.023	-0.362
	Moderately severe	0.146	2.050	-0.019	-0.271	0.105	1.606
	Very severe	0.336	3.909	0.019	0.228	0.066	0.795
	Extremely severe	0.436	4.312	0.264	2.472	0.182	1.746
Latent Constructs							
Factor score (continuous)	Social interaction propensity	--	--	--	--	0.066	2.217
	Community support capacity	0.067	2.368	0.069	2.482	--	--
	Personal resilience	0.070	2.447	0.091	3.146	-0.045	-1.526
Individual Characteristics							
Age (*)	51 to 65 years	--	--	--	--	-0.126	-2.323
	65 or older	-0.169	-2.894	-0.136	-2.233	--	--
Employment (Worker)	Non-worker	-0.263	-4.944	-0.205	-3.902	--	--
Disability (No)	Travel-related disability	0.082	1.399	--	--	--	--
Household Characteristics							
Income (*)	Under $50,000	--	--	--	--	-0.137	-2.179
	$50,000 to $100,000	--	--	--	--	-0.173	-2.671
Household size (3+)	1 person	--	--	-0.118	-1.952	-0.098	-1.613
	2 persons	-0.088	-1.809	-0.181	-3.171	-0.156	-2.823
Vehicles (1 or more)	Zero vehicles	0.139	1.916	--	--	--	--
Air conditioning (No)	Has A/C	0.104	1.565	--	--	--	--
Built Environment							
Area type (Non-rural)	Rural	-0.190	-2.149	--	--	--	--
Population density, people/acre (*)	High (>8.86)	--	--	0.112	2.043	--	--
Street network density, miles/sq. mile (*)	Medium (>0.40–1.11)	0.102	2.101	--	--	--	--
	High (>1.11)	--	--	-0.095	-1.869	--	--
Transit access (*)	High (>0.13)	0.153	2.115	0.132	1.666	--	--
Walkability index (*)	Low (>8–10)	--	--	--	--	-0.144	-2.224
Thresholds							
Cut-points	τ1 (Less | Same)	-1.002	-10.797	-1.047	-14.191	-0.948	-12.726
	τ2 (Same | More)	0.901	9.738	0.546	7.569	0.731	9.897
Model Fit							
Number of observations		2,653		2,434		2,381	
Log-likelihood at convergence		-2307.764		-2351.618		-2253.919	
Log-likelihood (thresholds only)		-2384.776		-2409.153		-2278.882	
McFadden pseudo-R^2		0.032		0.024		0.011	

Note: Ordered probit, one model per event (no intercept; the cut-points absorb it). Each model retains the four impact-severity dummies plus controls retained by backward elimination ($p < 0.10$). "--" = variable not in that event's final model; (*) base = all categories of the variable not listed. Latent constructs are standardized factor scores (Section 1). Positive coefficients shift responses toward the upper end of the scale. Cut-point t-statistics use delta-method standard errors. Built-environment categories: population density (people/acre): low ≤4.00, medium >4.00–8.86, high >8.86; street network density (miles/sq. mile): low ≤0.40, medium >0.40–1.11, high >1.11; transit access: low = 0, medium >0–0.13, high >0.13; National Walkability Index: very low ≤8, low >8–10, high >10–13, very high >13. Sample: all respondents exposed to each event.

3. AVERAGE TREATMENT EFFECTS

The Scenario Analysis page translates the estimated models into average treatment effects (ATEs): the change in the predicted share of respondents who would do less, about the same, or more of an activity when a scenario variable is changed and everything else is held fixed. This section gives the formulas behind the two panels of the page. Panel A changes the severity of the event's impact; Panel B changes a population characteristic, either categorical (B1) or continuous (B2). ATEs are computed from the estimated ordered probit models in Section 2; no model is re-estimated. Each ATE is computed within one event × activity model, on that model's estimation sample of N respondents, using that model's own coefficients and cut-points. A variable that is not in a model's final specification has a coefficient of zero in that model.

3.1. ORDERED PROBIT RESPONSE PROBABILITIES

For respondent i with covariate vector x_i, the linear index is $z_i = x_i'\beta$. The probability of response category j is

$$P_{ij}(z_i) = \Phi(\tau_j - z_i) - \Phi(\tau_{j-1} - z_i),\ j = 1, \ldots, J,\ \tau_0 = -\infty,\ \tau_J = +\infty \tag{1}$$

where $\Phi(\cdot)$ is the standard normal CDF and τ_j are the estimated cut-points. $J = 3$ for the activity-change outcomes (1 = Less, 2 = About the same, 3 = More), reported as *Do less*, *About the same* and *Do more*. For *Going about business as usual* ($J = 5$), the five likelihood categories are collapsed into the same three reported responses:

$$P_{i,less} = P_{i1} + P_{i2},\ P_{i,same} = P_{i3},\ P_{i,more} = P_{i4} + P_{i5} \tag{2}$$

i.e., Very/Somewhat unlikely → Do less, Neutral → About the same, Somewhat/Very likely → Do more.

3.2. GENERAL DEFINITION OF THE ATE

A scenario s fixes the manipulated variable(s) at chosen values for every respondent and leaves all other covariates at their observed values, giving an index $z_i^{(s)}$. The sample-average probability of response j under scenario s is

$$\overline{P}_j(s) = \frac{1}{N}\sum_{i=1}^{N} P_{ij}\left(z_i^{(s)}\right) \tag{3}$$

For a base scenario b and a comparison scenario c, the absolute ATE (Eq. 4) and the percentage ATE (Eq. 5) on response j are

$$ATE_j = \overline{P}_j(c) - \overline{P}_j(b) \tag{4}$$

$$ATE_j^{\%} = \frac{\overline{P}_j(c) - \overline{P}_j(b)}{\overline{P}_j(b)} \times 100 \tag{5}$$

The absolute ATE is a difference in probabilities (× 100 = percentage points); the percentage ATE is the change relative to the base-scenario probability. Because probabilities sum to one in each scenario, absolute ATEs sum to zero across the three responses. Averaging individual probabilities (rather than evaluating at the sample mean of x) keeps the non-linearity of $\Phi(\cdot)$ for every respondent.

3.3. PANEL A: EFFECT OF IMPACT SEVERITY

Let S = {imp_2, …, imp_5} denote the severity dummies of the event, with coefficients $\gamma_2, \ldots, \gamma_5$, and $\gamma_1 \equiv 0$ for the reference level (Not severe at all). Under scenario $l \in \{1, \ldots, 5\}$, every respondent is assigned severity level l, regardless of the observed severity, while all non-severity covariates stay at their observed values:

$$z_i^{(l)} = \sum_{k \notin S} \beta_k x_{ik} + \gamma_l,\ \gamma_1 = 0 \tag{6}$$

For a base level b and a comparison level c (any ordered pair, $b \neq c$; 20 pairs in total),

$$ATE_j(b \to c) = \frac{1}{N}\sum_{i=1}^{N}\left[P_{ij}\left(z_i^{(c)}\right) - P_{ij}\left(z_i^{(b)}\right)\right] \tag{7}$$

with Eq. (5) applied to the result for percentage ATEs. Since the same N respondents appear in both scenarios, the ATE isolates the effect of severity holding the sample composition fixed.

3.4. PANEL B: EFFECT OF POPULATION SEGMENTS

In Panel B, severity is **not** reset: each respondent keeps the observed severity level and all other observed covariates; only the segment variable is manipulated. The base and comparison scenarios for every contrast are listed in Table 20.

3.4.1. PANEL B1: CATEGORICAL VARIABLES

A categorical variable is represented by a set of dummies G (e.g., age: G = {age_3150, age_5165, age_65p}). A scenario is a profile $d = (d_k)_{k \in G}$ that sets all dummies in G jointly (e.g., 65 or older: d = (0, 0, 1); base 18–30: d = (0, 0, 0)). The index under profile d replaces the observed dummy values with d:

$$z_i^{(d)} = \sum_{k \notin G}\beta_k x_{ik} + \sum_{k \in G}\beta_k d_k = z_i + \sum_{k \in G}\beta_k\left(d_k - x_{ik}\right) \tag{8}$$

$$ATE_j\left(d^b \to d^c\right) = \frac{1}{N}\sum_{i=1}^{N}\left[P_{ij}\left(z_i^{(d^c)}\right) - P_{ij}\left(z_i^{(d^b)}\right)\right] \tag{9}$$

Dummies of G that are not in the model have $\beta_k = 0$. If none of the dummies whose values differ between d^b and d^c is in the model, the ATE is exactly zero and the contrast is flagged as “not in model” (e.g., 3+ persons → 1 person is zero for a model that contains hhsize2 but not hhsize1).

3.4.2. PANEL B2: CONTINUOUS VARIABLES (LATENT CONSTRUCTS)

For a latent construct $k \in$ {SE, CR, PR}, the base scenario keeps each respondent’s observed factor score x_{ik}, and the comparison scenario adds a one-unit increase, $\Delta = 1$, on the standardized factor-score scale (approximately one standard deviation):

$$ATE_j(\Delta) = \frac{1}{N}\sum_{i=1}^{N}\left[P_{ij}\left(z_i + \beta_k \Delta\right) - P_{ij}\left(z_i\right)\right], \Delta = 1 \tag{10}$$

The shift is a fixed additive unit for every model; it is neither a 1% change nor rescaled by the model-specific sample standard deviation. If construct k is not in the model ($\beta_k = 0$), the ATE is zero. Percentage ATEs again follow Eq. (5).

TABLE 20 Base and Comparison Scenarios for Panel B

Variable	Base scenario	Comparison scenario(s)	Model variables manipulated
Individual Characteristics			
Gender	Male	Female	*female*
Age	18–30 years	31–50; 51–65; 65 or older	*age_3150*, *age_5165*, *age_65p*
Race	Other race	White; Black; Asian	*white*, *black*, *asian*
Ethnicity	Non-Hispanic	Hispanic	*hispanic_c*
Education	No bachelor's degree	Bachelor's degree	*edu_bs*
Education	Below bachelor's	Bachelor's or higher	*bs_grad*
Employment	Worker	Non-worker	*non_wrkr*
Disability	No	Travel-related disability	*dis_yes*
Work setting	Not mainly outdoors	Works mainly outdoors	*work_out*
Telecommuting	Telecommutes	Does not telecommute	*tcom_no*
Household Characteristics			
Household income	Under $50,000	$50,000–$100,000; $100,000 or more	*in50*, *in50100*
Household size	3 or more persons	1 person; 2 persons	*hhsize1*, *hhsize2*
Children	No child	Has child(ren)	*child*
Vehicles	1 or more	Zero vehicles	*hhveh0*
Dwelling	Other dwelling	Stand-alone house	*sa_home*
Air conditioning	No A/C	Has A/C	*ac_c*
Built Environment			
Area type	Non-rural	Rural	*rural*
Population density (people/acre)	Low (≤4.00)	Medium (>4.00–8.86); High (>8.86)	*PopDens_medium*, *PopDens_high*
Employment density (jobs/acre)	Low (≤0.91)	Medium (>0.91–2.37); High (>2.37)	*EmpDens_medium*, *EmpDens_high*
Street network density (miles/sq. mile)	Low (≤0.40)	Medium (>0.40–1.11); High (>1.11)	*NetworkDensity_medium*, *NetworkDensity_high*
Land-use diversity	Low or medium (≤0.45)	High (>0.45)	*Diversity_high*
Transit access	Low (0)	Medium (>0–0.13); High (>0.13)	*TransitAccess_medium*, *TransitAccess_high*
National Walkability Index	Very low (≤8)	Low (>8–10); High (>10–13); Very high (>13)	*NatWalkInd_low*, *NatWalkInd_high*, *NatWalkInd_very_high*
Latent Constructs (continuous, Panel B2)			
Social interaction propensity	Observed score	Observed score + 1	*SE*
Community support capacity	Observed score	Observed score + 1	*CR*
Personal resilience	Observed score	Observed score + 1	*PR*

Note: Each comparison scenario is evaluated against the base scenario of the same row. Dummies of a variable not listed for a scenario are set to 0; e.g., for household income, Under $50,000 = (in50 = 1, in50100 = 0), $50,000–$100,000 = (0, 1), $100,000 or more = (0, 0). Severity is held at each respondent's observed level in all Panel B contrasts.